

To Verify Kirchhoff's Current Law

Start Time: 4:39:28 PM

End Time: 4:50:40 PM

Total Time Spent: 11 min 12 sec

Summary

Aim

To verify Kirchhoff's Current Law by measuring that the total current entering the junction is equal to the total current leaving it in a resistive DC circuit.

Simulation Summary

The guided walkthrough familiarised the user with the simulation's interface. The circuit was connected, and the connections were verified successfully. The resistance values were selected, and the DC supply voltage was varied to measure the branch currents at different voltage values. The ammeter readings were recorded, and the measured currents were used to verify Kirchhoff's Current Law (KCL) by confirming that the total current entering a junction is equal to the total current leaving the junction. Along with the KCL verification, the current v/s voltage graph was also plotted using the measured readings.

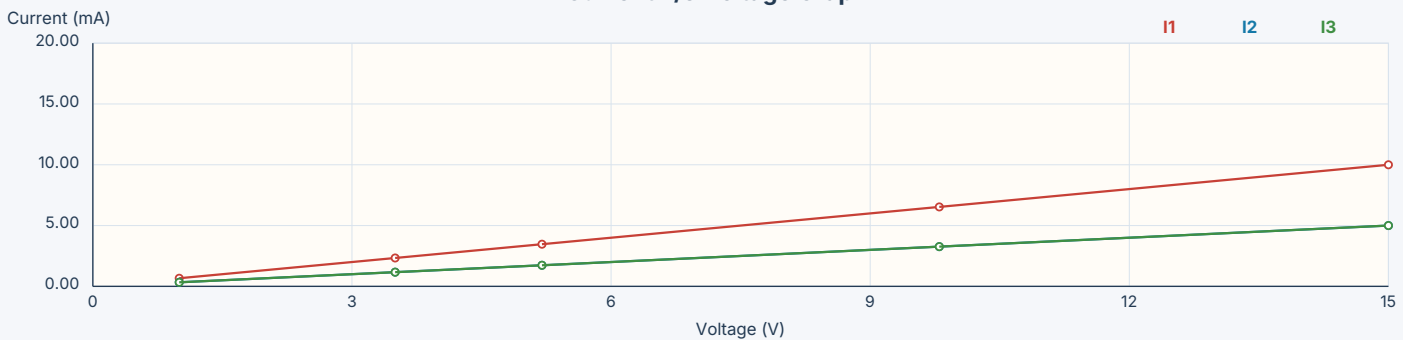
Components and Key Parameters

- DC power supply: 15 V
- DC Ammeter A1 for total current I1: 0 - 10 mA
- DC Ammeter A2 for branch current I2: 0 - 10 mA
- DC Ammeter A3 for branch current I3: 0 - 10 mA
- R1: 1 k Ω
- R2: 1 k Ω
- R3: 1 k Ω
- Connecting leads

Observation Table

S.No.	Voltage (V)	I1 (mA)	I2 (mA)	I3 (mA)
1	3.50	2.33	1.17	1.17
2	1.00	0.67	0.33	0.33
3	5.20	3.47	1.73	1.73
4	9.80	6.53	3.27	3.27
5	15.00	10.00	5.00	5.00

Current v/s Voltage Graph



Theoretical Verification

S.No	Verified Reading	R (k Ω)	I1 (mA)	I2 (mA)	I3 (mA)	Status
1	Reading 1 (3.50 V)	1.50	2.33	1.17	1.17	Verified
2	Reading 2 (1.00 V)	1.50	0.67	0.33	0.33	Verified
3	Reading 3 (5.20 V)	1.50	3.47	1.73	1.73	Verified
4	Reading 4 (9.80 V)	1.50	6.53	3.27	3.27	Verified
5	Reading 5 (15.00 V)	1.50	10.00	5.00	5.00	Verified

Conclusion

For each recorded voltage value, the total current I1 was found to be equal to the sum of branch currents I2 and I3. Moreover, the theoretically calculated currents were found to be the same as the recorded readings. Hence, Kirchhoff's Current Law was successfully verified for the given resistive DC circuit.